Task F: backpropagation

Given

Sample

$x = 1$, target $y = 2$

Model

$h = \tan(wx + b)$

$\hat{y} = vh + c$

Loss: $J = \frac{1}{2}(\hat{y} - y)^2$

Start: $w = 0.5$, $b = 0$, $v = 1$, $c = 0$

Learning rate: $\alpha = 0.1$

① Forward pass

$z = wx + b = 0.5(1) + 0 = 0.5$

$h = \tan(0.5) = 0.462117$

$\hat{y} = vh + c = 1(0.462117) + 0 = 0.462117$

error $= \hat{y} - y = 0.462117 - 2 = -1.537883$

$J = \frac{1}{2}(-1.537883)^2 = 1.182542$

② Backward pass (chain rule)

$$\frac{\partial J}{\partial \hat{y}} = \hat{y} - y$$

$$\frac{\partial \hat{y}}{\partial v} = h, \quad \frac{\partial \hat{y}}{\partial c} = 1, \quad \frac{\partial \hat{y}}{\partial h} = v$$

$$\frac{\partial h}{\partial z} = 1 - \tan^2 z = 1 - h^2$$

$$\frac{\partial z}{\partial w} = x \qquad \frac{\partial z}{\partial b} = 1$$

Output layer

$\frac{\partial J}{\partial c} = (\hat{y} - y) \cdot 1 = -1.537883$

$\frac{\partial J}{\partial v} = (\hat{y} - y) \cdot h = -0.710682$

Hidden layer

$1 - h^2 = 1 - (0.462117)^2 = 0.786448$

$\partial J/\partial z = (\hat{y} - y) \cdot v \cdot (1 - h^2) = (-1.537883)(1)(0.786448)$

$= -1.209464$

$\partial J/\partial w = \partial J/\partial z \cdot x = (-1.209464) = -1.209464$

$\partial J/\partial b = \partial J/\partial z \cdot 1 = -1.209464$

One gradient descent step

Rule $\theta_{new} = \theta - \eta \cdot \frac{\partial J}{\partial \theta}$

$w_{new} = 0.5 - 0.1(-1.209464) = 0.620946$

$b_{new} = 0 - 0.1(-1.209464) = 0.120946$

$v_{new} = 1 - 0.1(-0.710682) = 1.071068$

$c_{new} = 0 - 0.1(-1.537883) = 0.153788$

New Parameters

$z = 0.620946(0) + 0.120946 = 0.741893$

$h = \tanh(0.741893) = 0.630287$

$t = 1.071068(0.630287) + 0.153788 = 0.828869$

$J = 1/2 (0.828869 - 2)^2 = 0.685774$

→ Loss from $1.182542 \rightarrow 0.685774$